Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/21/2006
PI Name:	Sullivan, Donald B. Ph.D.		
Project Title:	Primary Atomic Reference Clock in Space (PARCS)		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	303-497-3276
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	NIST MS 847.00		
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PI Web Page:	http://www.boulder.nist.gov/		
City:	Boulder	**State:**	CO
Zip Code:	80305-3328	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	05/01/1998	**End Date:**	03/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
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Flight Program:			
Flight Assignment:	will most likely get an extension into March 2007, per correspondence 7/06.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donley, Elizabeth (NIST) Heavner, Tom (NIST) Hollberg, Leo (NIST) Jefferts, Steve (NIST) Levine, Judah (NIST) Parker, Tom (NIST) Phillips, Bill (NIST) Robinson, Hugh (NIST) Walls, Fred (NIST) Ashby, Neil (University of Colorado) Dick, John (JPL) Klipstein, Bill (JPL) Maleki, Lute (JPL) Purcell, George (JPL) Seidel, Dave (JPL) Thompson, Rob (JPL) Wu, Sien (JPL)		

	Young, Larry (JPL) Mattison, Ed (Harvard SAO) Vessot, Robert (Harvard SAO) De Marchi, Andrea (Politecnico di Torino)
Grant/Contract No.:	C73537A
Performance Goal No.:	1.24
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity
Task Description:	The program involves the development of a laser-cooled cesium-beam atomic clock and related systems for flight on the International Space Station (ISS). The clock is designed to perform certain tests of gravitational theory and to serve as a primary frequency standard. The key advantage of putting an atomic clock in space is the absence of gravity, which limits the performance of these clocks on Earth. The limitation on Earth relates to the fact that atoms cannot remain in the clock for more than about 1 second, the time it takes to loft a ball of atoms about 1 meter and observe the atoms as they rise and then fall under the influence of gravity. In the microgravity of space, the observation time for atoms can be extended to times as long as perhaps 10 seconds. The effect of this is that the resonance defining the length of the second becomes more sharply defined, and certain perturbing effects that lead to inaccuracy are reduced. As part of NASA's realignment to new tasks, the Fundamental Physics Program, along with PARCS and all other efforts within the program, has been cancelled. Efforts in this year involve capturing the advances that have been made and publishing a paper describing the status of PARCS. JPL served as the leader in the development of all flight hardware; the University of Colorado led the theoretical and analytical studies relating to the gravitational/relativity studies; and NIST worked on concept development and specific design requrements, and planning of the sequence of experiments and analysis methods to be used during flight. The scientific and technical objectives of the program were: 1. to search for possible deviations from theory of the expected dependence of the frequency shift in clocks on gravitational potential; 2. in conjunction with SUMO (Superconducting Microwave Oscillator), which is scheduled to fly concurrently, to perform a Kennedy-Thorndike test, which looks at the spatial isotropy of the speed of light; 3. to search for possible violations of the local-position-invariance principle that indicates that clocks based on different atoms/principles should continue to run at the same relative rate as the gravitational potential is varied; 3. to study the behavior of laser-cooled atoms in microgravity, where they can be observed for substantially longer periods than on earth; 4. to improve the realization of the second and demonstrate that this improvement can be widely distributed to earth from the ISS; and 5. to study Global Positioning System satellite signals from above the troposphere and ionosphere in order to augment ground-collected information on GPS clock performance. The significance of this experiment lies in several distinctly different areas. First, high-performance atomic clocks in space provide a direct means for testing gravitational theory, and while this particular experiment will lead to certain immediate tests of theory, it will also serve as the test bed for proving the clock concepts, which in turn will lead to even more ambitious clock-based gravitational tests, particularly in flights involving very large changes of gravitational potential. Second, since the gravitational field of the earth is now limiting the performance of earth-bound atomic clocks, this microgravity experiment provides the opportunity to improve upon these clocks and could eventually lead to a new era where time/frequency standards reside in space. Third, a substantial amount of modeling of the behavior of atoms in microgravity has been done, and this will be the first opportunity to test these models. Finally, the experiment provides the opportunity to compare the performance of atomic clocks aboard GPS satellites with this highly stable atomic clock without the complication of delay variations associated with transmission of comparison signals through the troposphere and ionosphere. Such studies could improve our understanding of GPS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The program has resulted in a number of advances in atomic-clock design, many of which are useful for terrestrial atomic clocks. Such advances include vacuum shutters for interrupting atomic beams, phase modulation for interrogation of the resonance, improved laser/optical systems, and new off-resonant cavity designs.
Task Progress:	A paper entitled "PARCS: NASA's Laser-Cooled Atomic Clock in Space" was presented at the 35th COSPAR Scientific Assembly in Paris and published in the conference proceedings (December 2005). Among other things, this paper focused on the distinctions between the designs of the PARCS and ACES experiments. The more ambitious PARCS program involved innovations that would have allowed its performance to exceed that of atomic clocks on Earth. The ACES design cannot do that.
Bibliography Type:	Description: (Last Updated: 07/21/2006)
Articles in Peer-reviewed Journals	Sullivan DB, Ahsby N, Donley EA, Heavner TP, Hollberg SR, Jefferts SR, Klipstein WM, Phillips WD, Seidel DJ. "PARCS: NASA's laser-cooled atomic clock in space." Adv Space Res. 2005;36(1):107-13. , Dec-2005